

Learning with confidence

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Abstract

This report proposes a creativity-oriented variant of GRPO for large language models. The key idea is that modern reasoning models are often extremely confident on mathematical tasks, and that this confidence can reflect a narrow, well-rehearsed response manifold rather than flexible reasoning. The modified objective replaces the standard KL penalty with entropy regularization and rewards lower-confidence but correct outputs over highly confident responses that may reflect memorization.

1 Introduction

Recent reinforcement learning methods for reasoning-oriented language models aim to improve answer quality without relying entirely on supervised imitation. Among these methods, GRPO is attractive because it avoids the explicit critic or value network used in PPO-style methods and instead uses within-group reward normalization as a baseline [1, 2]. This improves memory efficiency, but the standard objective still inherits an important structural bias: it constrains the learned policy to remain close to a reference model. For tasks where creativity, non-standard reasoning, or genuine exploration are desirable, this is a serious limitation.

This project begins from a simple observation about strong reasoning models on math tasks: when they succeed, they often do so with very high confidence. That is desirable for reliability, but it can also indicate that the model is moving through familiar solution patterns rather than exploring alternative ways of solving the problem. If the training objective continues to reward the policy for concentrating probability mass around already likely answers, then stronger reasoning performance may come at the cost of narrowing the space of possible reasoning strategies.

The research goal of this project is to redesign GRPO so that the policy is not rewarded merely for reproducing highly probable responses, but instead for discovering lower-confidence yet correct reasoning trajectories. The underlying thesis is that confident correctness often reflects memorized or highly familiar patterns, whereas correctness reached under lower answer confidence can be evidence of more original inference. But at the same time assuming that the model is still confident over locked pattern of its output.

This motivation is particularly important for models that are already heavily post-trained with instruction tuning and reinforcement learning from feedback. Such models are typically optimized to be helpful, aligned, and reliable, but that same optimization pressure can narrow the policy

around polished, high-confidence completions that resemble previously rewarded behavior. In that setting, a creativity-oriented GRPO objective is useful because it counteracts over-consolidation: rather than further rewarding the model for staying near its existing response manifold, it creates room for less rehearsed reasoning paths that are still grounded by correctness.

2 Background and Motivation

In standard GRPO, a model samples a group of outputs for a prompt and uses the relative distribution of rewards within that group to estimate the advantage. This removes the need for a separate value model, but the optimization still contains a KL penalty to a frozen reference policy [2]. The baseline objective is

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\sum_{i=1}^G \min \left(\frac{\pi_{\theta}(o_i | q)}{\pi_{\theta_{\text{old}}}(o_i | q)} \hat{A}_i, \text{clip} \left(\frac{\pi_{\theta}(o_i | q)}{\pi_{\theta_{\text{old}}}(o_i | q)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_i \right) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right],$$

with group-normalized advantage

$$\hat{A}_i = \frac{r_i - \mu_r}{\sigma_r}.$$

The major limitation is architectural rather than purely empirical. The KL term explicitly discourages movement away from the reference model distribution. If the reference model encodes dominant training-set patterns, then the learning signal systematically punishes novel trajectories even when they are correct. For a creativity-oriented objective, tuning the KL coefficient is therefore not enough; the anchoring mechanism itself must be removed.

Two additional observations motivate the modified design. First, confidence is not equivalent to correctness, because language models can be confidently wrong. Second, the most meaningful confidence signal is not the probability of the full generated trace, but the sequence probability over the final answer span only. Reasoning tokens are often verbose and noisy, so using them directly can distort reward shaping. The core idea of this report is therefore deliberately modest: rather than trying to define creativity directly, we treat lower answer-span confidence on correct outputs as a measurable proxy for reasoning that is less purely imitative.

Recent OpenAI safety work provides an important adjacent motivation for this project. Their results suggest that current reasoning models often struggle to deliberately control or obfuscate their chain-of-thought, which makes those reasoning traces useful for monitoring. At the same time, OpenAI reports that directly optimizing the chain-of-thought to satisfy external constraints can encourage models to hide intent rather than eliminate undesirable behavior. Taken together, these findings support a design strategy in which training should shape answer quality and exploration incentives without naively treating the visible chain-of-thought itself as a fully controllable or fully reliable optimization target [4, 5].

Related work from Anthropic sharpens the same concern from a different angle. Their studies on alignment faking and sleeper agents suggest that models can preserve undesirable objectives,

strategically behave differently under monitoring, and in some cases retain deceptive behaviors even after additional safety training. This strengthens the case for objectives that do not confuse surface compliance with genuine internal generalization. In the context of this report, it further motivates measuring creativity and correctness through robust behavioral signals, rather than assuming that polished responses or apparently cooperative reasoning traces reflect the true underlying policy [6, 7].

3 Modified Objective

The proposed objective replaces the reference-model KL penalty with an entropy regularizer and introduces a creativity-oriented advantage definition:

$$\mathcal{L}^*(\theta) = \mathbb{E} \left[\sum_{i=1}^G \min \left(\rho_i \hat{A}_i^*, \text{clip}(\rho_i, 1 - \epsilon, 1 + \epsilon) \hat{A}_i^* \right) + \lambda \mathcal{R}_H(\pi_\theta) \right],$$

where

$$\rho_i = \frac{\pi_\theta(o_i | q)}{\pi_{\theta_{\text{old}}}(o_i | q)} \quad \text{and} \quad \hat{A}_i^* = s_i \cdot (1 + r_i).$$

Here, r_i is the correctness reward and $s_i \in \{-1, +1\}$ is a sign function that changes the incentive structure based on answer-span confidence. The sign is defined as

$$s_i = \begin{cases} -1 & \text{if } r_i > 0 \text{ and } p_i > \tau, \\ +1 & \text{if } r_i > 0 \text{ and } p_i \leq \tau, \\ -1 & \text{if } r_i \leq 0, \end{cases}$$

where p_i is the sequence probability of the answer span and τ is a confidence threshold. This design treats confident-correct answers as potential “parrot behavior” and lower-confidence correct answers as evidence of productive exploration.

The answer-span confidence is computed as

$$p_i = \exp \left(\frac{1}{|\mathcal{A}_i|} \sum_{t \in \mathcal{A}_i} \log \pi_\theta(o_t | o_{<t}, q) \right),$$

where $\mathcal{A}_i$ indexes only tokens in the answer span. This excludes the reasoning span from confidence estimation and focuses the signal on the final committed prediction.

The entropy regularizer is

$$\mathcal{R}_H(\pi_\theta) = \frac{1}{T} \sum_{t=1}^T H(\pi_\theta(\cdot | o_{<t}, q)),$$

with token-level entropy

$$H(\pi_\theta(\cdot \mid o_{<t}, q)) = - \sum_{v \in \mathcal{V}} \pi_\theta(v \mid o_{<t}, q) \log \pi_\theta(v \mid o_{<t}, q).$$

Unlike the standard KL penalty, this term does not tether the policy to a known distribution; instead, it rewards maintaining useful uncertainty during generation.

4 Implementation Status

The current prototype is implemented in `creative_grpo.py`. The implemented components include answer-span sequence log-probability computation, the sign function s_i , the modified advantage computation, PPO-style clipped optimization using $\hat{A}_i^*$, and the entropy regularizer $\mathcal{R}_H$. The optimization structure follows the general PPO-style surrogate formulation while replacing the standard GRPO anchoring term [1, 2]. The code for this project is available at the-deep-field/Learning_with_confidence for reference and continued development.

Initial smoke tests confirm that the sign and advantage calculations behave as expected across alternating correct and incorrect reward examples. This is only a preliminary functional check, but it provides evidence that the incentive mechanism is wired correctly before full-scale training experiments.

5 Lower Confidence as a Measure

The main conceptual bet of this paper is that lower confidence itself can serve as a usable measure when it appears alongside correctness. The goal is not to claim that every uncertain answer is creative, nor that every confident answer is rote. Instead, the claim is narrower: among correct solutions, those that the model assigns lower answer-span probability may reveal reasoning trajectories that are less overlearned and less tightly compressed into the policy.

This framing keeps the method operationally simple. It avoids introducing a separate novelty score that would require its own modeling assumptions and could easily collapse into measuring randomness. By contrast, answer-span confidence is already available from the policy, can be computed per example, and fits naturally into both optimization and evaluation.

The open question is therefore empirical rather than definitional: whether correct low-confidence outputs truly correlate with more interesting or more transferable reasoning. The proposed objective is designed as a clean way to run that experiment.

6 Design Principles

The project is guided by five principles. First, KL divergence is viewed as architecturally anti-creative, so it should be replaced rather than merely reweighted. Second, confidence must not be

conflated with correctness, and the objective should exploit the gap between them. Third, answer-span probability is the appropriate confidence signal because reasoning spans are too noisy. Fourth, lower confidence on correct answers is treated as a practical proxy for less rehearsed reasoning. Fifth, GRPO’s within-group normalization remains useful because it supplies a baseline without a separate value model and keeps the objective operationally measurable and auditable directly [2, 3].

7 Conclusion

This work proposes a principled modification of GRPO aimed at encouraging creative reasoning in large language models. The method removes reference-model anchoring and rewards lower-confidence but correct outcomes. Rather than trying to engineer a separate novelty score, the proposal treats answer-span uncertainty itself as the central measurable signal. The most important next step is to test whether this signal actually distinguishes memorized behavior from more transferable reasoning. If it does, the proposed framework could provide a concrete route toward reinforcement learning objectives that optimize not only for correctness, but also for more genuinely exploratory reasoning.

8 References

References

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